One split at a time is often enough: empirical limits of bounded lookahead in decision-tree induction

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Abstract

Decision trees are usually induced by greedy recursive splitting, although a split's value depends on the subtree it permits. Multi-step lookahead is a natural compromise between greedy induction and exact optimal-tree search: it preserves the recursive tree-growing paradigm but scores candidate splits through their descendants. We study whether this additional non-greedy optimization materially improves shallow decision trees and random forests when predictive accuracy and fitting time are evaluated jointly. On a sample of 67 PMLB classification datasets, lookahead depths 2 and 3 raised mean accuracy over ordinary depth-1 greedy induction, but only by small and uneven margins. For single trees, mean accuracy increased from 0.6931 to 0.7100 and 0.7145, while mean fitting time increased from 0.008 s to 0.373 s and 70.90 s. For forests, mean accuracy increased from 0.7068 to 0.7125 and 0.7260, while fitting time increased from 0.019 s to 0.694 s and 96.19 s. A simple accuracy-time utility calculation makes the trade-off explicit: over the tested depths, modest accuracy gains were accompanied by orders-of-magnitude increases in fitting time. In a completed five-dataset forest-size pilot, adding greedy trees was more effective than making each tree farsighted. These results suggest that bounded lookahead is better viewed as a targeted tool than as a default replacement for greedy CART-style induction.

Keywords: Decision trees, Random forests, Lookahead search, Greedy algorithms, Optimal trees, PMLB

1 Introduction

The standard decision-tree algorithm makes a sequence of local decisions. At each internal node it selects the split that most improves an impurity criterion, then repeats the same operation on the two child nodes [1, 2]. This strategy is fast, simple, and effective enough to remain the foundation of single decision trees, bagged trees, and random forests [3, 4]. Its weakness is equally clear: a split is not useful in isolation, but because of the subtree it enables. A locally weak split may create children that are easy to separate, while a locally strong split may make subsequent separation difficult.

This paper studies the practical value of relaxing that greediness. An n-step lookahead tree scores a candidate split by looking n generations ahead, choosing the split whose best descendant subtree gives the largest impurity reduction. The method is appealing because it occupies a middle ground: it is less myopic than CART-style induction, but far less ambitious than exact global optimization of the whole tree.

The research question is deliberately narrow: does bounded multi-step split optimization deliver enough predictive accuracy to justify its computational cost? We answer this question empirically for shallow single trees and forests. The central finding is that lookahead can improve mean accuracy, but the improvement is small and uneven compared with the increase in fitting time. The result is not that greedy trees are optimal; they are not. It is that greedy induction remains highly competitive once time is part of the objective.

2 Related work

Top-down induction of decision trees was codified in ID3, CART, and related algorithms as a greedy search over split scores or impurity reductions [1, 2]. Bagging and random forests changed the statistical behavior of greedy trees by averaging many unstable predictors [3, 4], but they did not remove the greedy split-selection mechanism inside each tree. This matters for the present study because lookahead inside a forest competes not only with a single greedy tree, but also with the error-reduction effect of ensembling.

Lookahead tree induction has a long history. Norton [5] proposed generating better decision trees by considering downstream tree quality, and Ragavan and Rendell [6] studied lookahead feature construction for difficult concepts. Murthy and Salzberg [7] gave a particularly relevant warning: one-level lookahead can be much more expensive and does not reliably improve accuracy, a phenomenon they connected to pathology in decision-tree induction. Elomaa and Malinen [8] further analyzed lookahead and pathology. Esmeir and Markovitch [9] reframed tree induction as an anytime process, where additional computation should be justified by measurable improvement in the learned tree. Our work follows this cost-sensitive perspective.

At the other end of the spectrum, exact or near-exact optimal-tree methods search directly over tree structures. Hyafil and Rivest [10] showed that constructing optimal binary decision trees is NP-complete. Modern mixed-integer and branch-and-bound approaches have made optimal trees practical in selected regimes [11–15], but scalability remains central. Other non-greedy methods optimize tree parameters jointly or through surrogate objectives [16]. Bounded lookahead is therefore an intermediate case: it asks whether a small amount of structured non-greediness captures enough of the available gain to be worthwhile in routine induction.

3 Lookahead induction and an accuracy-time utility

Let S be the samples reaching a node, let $I(S)$ be the node impurity, and let $\mathcal{T}_n(s, S)$ denote the frontier leaves obtained after applying candidate split s at the current node and then recursively choosing descendant splits for $n - 1$ additional generations. A depth-n lookahead split maximizes

$$G_n(s; S) = I(S) - \sum_{L \in \mathcal{T}_n(s,S)} \frac{|L|}{|S|} I(L). \tag{1}$$

Depth $n = 1$ is ordinary greedy induction. Depths $n = 2$ and $n = 3$ choose a split using the best two- or three-generation subtree available under the same candidate-split mechanism.

Increasing n changes the local training-set criterion from immediate impurity decrease to impurity after a bounded descendant expansion. This may correct a myopic split, but it need not improve held-out accuracy or preserve the greedy ranking of root splits. The difficulty is computation. If m candidate splits are considered at each internal node, the number of split configurations in a full binary lookahead subtree has a naive upper bound of order $m^{2^n - 1}$. Practical implementations reuse work and stop at pure or infeasible nodes, but the qualitative pressure remains: lookahead depth increases the search cost much faster than it increases the number of fitted tree levels.

To compare accuracy with time, we use a simple descriptive utility heuristic. Suppose mean predictive accuracy and mean fitting time at lookahead depth n are approximated by

$$A(n) \approx A(1) + \alpha(n - 1), \qquad T(n) \approx T(1) \exp\{\beta(n - 1)\}. \tag{2}$$

For a user who values accuracy but penalizes multiplicative increases in fitting time, define

$$U_\lambda(n) = A(n) - \lambda \log \left(\frac{T(n)}{T(1)} \right), \tag{3}$$

where λ is the accuracy cost assigned to one natural-log unit of additional time. Relative to the greedy baseline, a deeper lookahead depth has higher utility only if

$$A(n) - A(1) > \lambda \log \left(\frac{T(n)}{T(1)} \right). \tag{4}$$

Thus the empirical quantity

$$\lambda_n^\star = \frac{A(n) - A(1)}{\log(T(n)/T(1))}$$

is a break-even value: if the chosen time penalty exceeds $\lambda_n^\star$, greedy induction has higher utility under this model. The calculation is not intended as a full decision-theoretic model; it is a compact way to put accuracy and fitting time on the same scale.

Table 1 Aggregate performance over 67 PMLB classification datasets.

Estimator	Lookahead	Mean accuracy	Median accuracy	Mean fit time (s)
Decision tree	1	0.6931	0.7554	0.0076
Decision tree	2	0.7100	0.7222	0.3727
Decision tree	3	0.7145	0.7333	70.9025
Random forest	1	0.7068	0.7500	0.0188
Random forest	2	0.7125	0.7407	0.6940
Random forest	3	0.7260	0.7612	96.1864

4 Experiments

We evaluated lookahead depths $n \in \{1, 2, 3\}$ for a decision-tree classifier and a random-forest classifier on a sample of 67 PMLB classification datasets [17]. The experimental protocol was fixed throughout: a single train-test split, stratified when possible, test size 0.25, random state 0, maximum tree depth 3, all features available at each split, and no explicit cap on split candidates. The forest experiments used 100 bootstrap trees with the same lookahead rule inside each tree, so the comparison isolates the split-selection rule rather than feature subsampling.

We recorded test accuracy and wall-clock fitting time. Wall-clock times are implementation- and hardware-dependent, but their order of magnitude is relevant because lookahead changes the fitting problem itself. The goal is not to prove that lookahead never helps. Rather, it is to measure whether a routine move from one-step to multi-step split optimization changes the practical accuracy-cost frontier for shallow trees and forests under a fixed screening protocol.

5 Results

Table 1 summarizes the main benchmark. Lookahead improves mean accuracy in both model families, but the gain is modest compared with the cost. For single trees, lookahead depth 2 improves mean accuracy by 1.69 percentage points over depth 1, while mean fitting time is about 49 times larger. Lookahead depth 3 improves mean accuracy by 2.15 percentage points, while mean fitting time is about 9.4×10^3 times larger. For forests, depth 2 improves mean accuracy by 0.57 percentage points, while mean fitting time is about 37 times larger. Depth 3 improves mean accuracy by 1.92 percentage points, while mean fitting time is about 5.1×10^3 times larger.

Figure 1 displays the empirical pattern. Accuracy gains are approximately linear, and remain modest, over the three lookahead depths. Fitting time grows by orders of magnitude on the same grid, consistent with the exponential-time side of the heuristic above. This mismatch is the central empirical result: the lookahead signal is visible, but the computational price rises much more sharply.

The utility heuristic in Eq. (3) makes the tradeoff explicit. The break-even values $\lambda_n^\star$ are 0.00435 for tree lookahead depth 2 versus 1, 0.00235 for tree lookahead depth 3 versus 1, 0.00158 for forest lookahead depth 2 versus 1, and 0.00225 for forest lookahead depth 3 versus 1. These values correspond to roughly 0.16–0.44 percentage points of

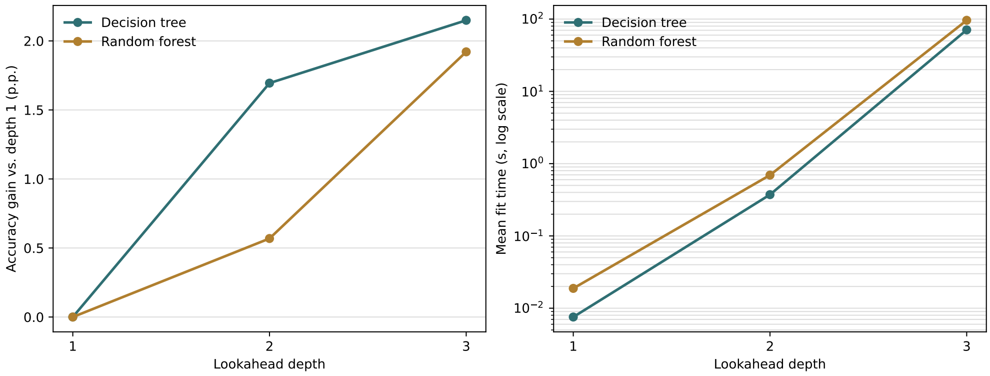

Fig. 1 Mean accuracy gain and mean fitting time as lookahead depth increases from 1 to 3. Accuracy gains are measured relative to the depth-1 greedy baseline. Fitting time is shown on a logarithmic scale.

accuracy per natural-log unit of additional fitting time. A user who charges more than that for training-time growth would prefer the greedy baseline in these aggregate comparisons.

The gains are also not uniform. Counting ties as wins, so that more than one depth may be credited on the same dataset, depth 1 was tied for the best single-tree accuracy on 30 datasets, depth 2 on 31 datasets, and depth 3 on 37 datasets. For forests, the corresponding counts were 36, 29, and 29. The median results reinforce the same point: depth 2 has a higher mean than depth 1 in both model families, but a lower median accuracy. Deeper lookahead therefore changes the average, but it does not dominate the greedy rule across datasets.

5.1 Can more greedy trees replace fewer lookahead trees?

We also examined the completed five-dataset subset from a smaller forest-size pilot. The question was whether adding more depth-1 trees can compete with using fewer depth-2 trees. Figure 2 shows the result. In this completed subset, larger greedy forests were at least as accurate as the depth-2 forests and were much faster. The best mean accuracy among the pilot configurations was obtained by 100 greedy trees (0.8721 mean accuracy, 0.187 s mean fit time). The best depth-2 configuration used 10 trees (0.8557 mean accuracy, 3.744 s mean fit time). This pilot is not a replacement for a full forest-size benchmark, but it suggests a practical hypothesis: when computation is fixed, increasing the number of greedy trees may often be preferable to making each tree farsighted.

6 Discussion

The empirical result is sharp but limited: bounded lookahead is a valid correction to greedy myopia, but it is not a compelling default under this protocol. Its optimization advantage is real, and the benchmark confirms that deeper lookahead can raise mean accuracy. Yet the improvement is thin relative to the cost, and it is not consistent across datasets.

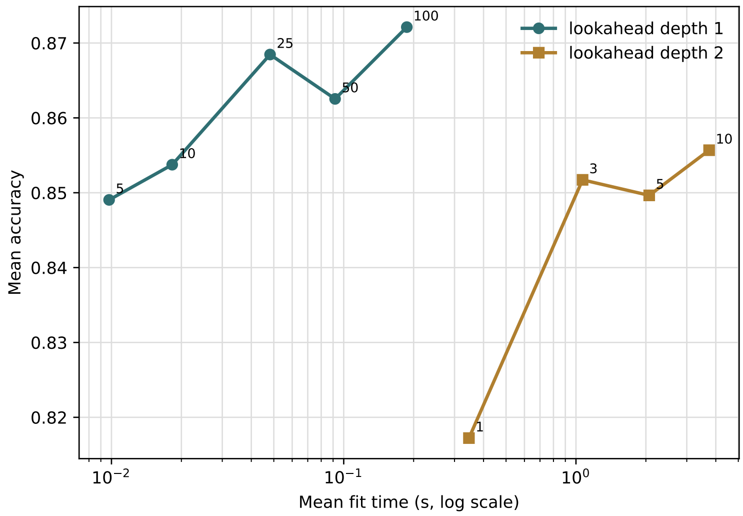

Fig. 2 Pilot comparison of forest size and lookahead depth on five small PMLB datasets with a completed smaller forest-size grid. Labels indicate the number of trees. In this pilot, larger greedy forests dominate the depth-2 configurations on the reported mean accuracy-time frontier.

This conclusion helps explain the durability of greedy decision trees. Greedy splitting is not merely a historical convenience; it gives a strong accuracy-time compromise. Random forests further reduce the cost of myopic split decisions by averaging many greedy trees. Once this ensemble mechanism is available, spending large amounts of computation inside each split decision has to clear a high bar. The forest-size pilot points in the same direction, although only as limited supporting evidence from the completed small-dataset runs.

The utility model is intentionally simple, but useful. When accuracy gains remain incremental while time grows by orders of magnitude, the marginal utility of deeper lookahead must eventually collapse for any positive time penalty. This does not rule out lookahead. It identifies where lookahead should be used: small problems, shallow trees, specialized interaction structures, interpretability settings where a single stronger tree is valuable, or cases where computation is cheap relative to the cost of prediction errors. The present evidence should be read with the same restraint: the protocol uses shallow trees, a fixed train-test split, and mean accuracy-time summaries rather than a full model-selection study.

7 Conclusion

Multi-step split optimization can improve decision trees and random forests, but in these experiments it does so by a narrow margin and at a very large computational cost. The practical lesson is therefore conservative. Greedy CART-style induction is not globally optimal, but it remains a remarkably strong default. For routine shallow tabular trees and forests, one split at a time is often the better starting point.

Data availability. The benchmark datasets are available from PMLB [17]. Aggregate tables, plotting scripts, and the implementation used for this study are included in the accompanying repository branch.

Code availability. The lookahead-tree implementation and manuscript-generation scripts are included in the accompanying repository branch.

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